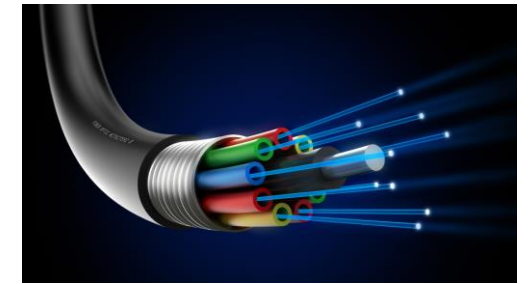
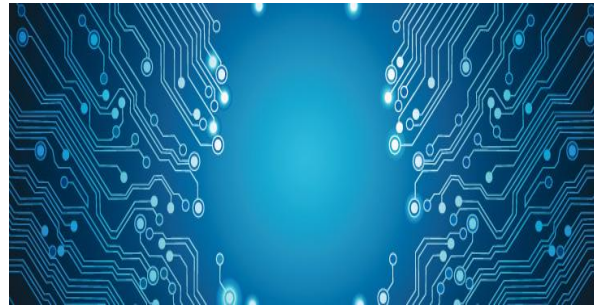


NanoXplore – Unleashing A New Generation of Graphene-Polymer Composites

NAIHENG SONG, PH.D.

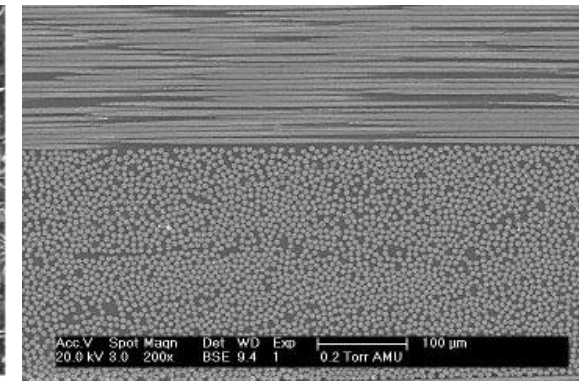
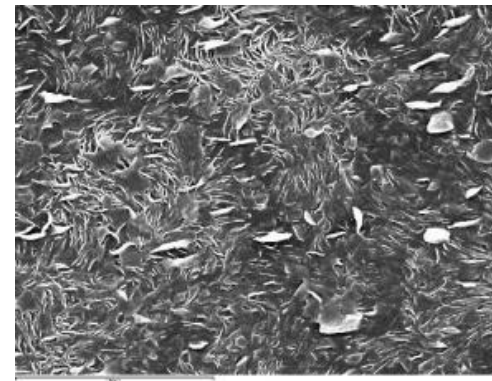
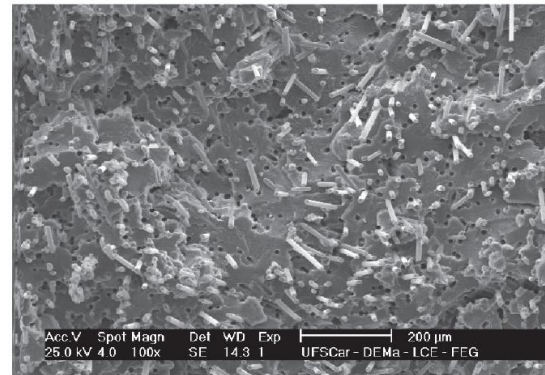
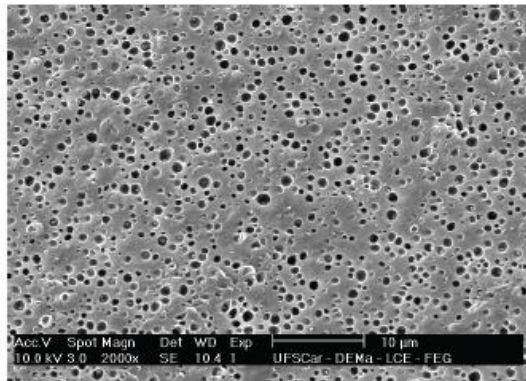
NOV 19, 2015

Towards Polymeric Materials with Ever-Improving Properties



- Polymer science has developed rapidly in the past century; polymeric materials are now widely used in our daily life
- Properties of polymers are being pushed to a plateau; however, ever-improving properties and functionalities are demanded
- Molecular/structural engineering and polymer-based composite materials are being actively pursued

Polymer-Based Composite Materials



- Materials of distinctly different types are combined to achieve enhanced properties;
- Easy to prepare and flexible in modification according to applications
- Challenges: typically a significant amount of “second component” has to be added, which very often imposes some negative effect on the base polymer, such as processability, mechanical properties, electrical properties, etc.

Towards A New Generation of Polymer Composite Materials

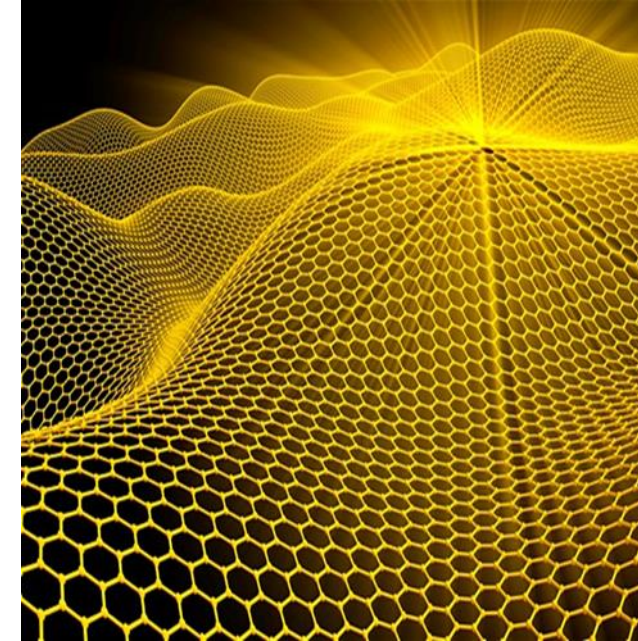
Polymer-based composite materials with significantly enhanced certain selected properties without sacrificing other desirable properties of pristine polymers

- Mechanical strength vs Toughness & Processability
- Thermal conductivity vs Workability & Electrical properties
- Thermal conductivity vs optical properties
- Charge transporting properties vs optical properties
- Aesthetics

Graphene – The Enabler for the New Generation of Polymer Composites

Graphene holds the answer for the next generation of polymer composites

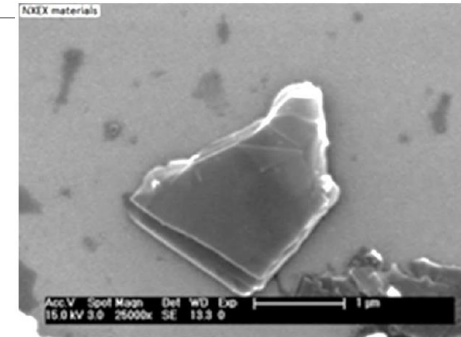
- Tensile strength – 130 Gpa
 - Young's modulus – 1 Tpa
 - Thermal conductivity – $5300 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$
 - Electron/hole mobility – $15,000 \text{ cm}^2\cdot\text{V}^{-1}\cdot\text{s}^{-1}$
 - Electrical resistivity – $10^{-6} \Omega\cdot\text{cm}$
 - Surface area – $2630 \text{ m}^2\cdot\text{g}^{-1}$
 - Bandgap – 0~0.25 eV
-
- Potential applications: electronics, photovoltaics, energy storage, biological engineering, composite materials, etc.
 - Challenges – Practical application of graphene, e.g., in form of polymer composites - quality, dispersibility, cost-effectiveness, customer's acceptance to the new technology



NanoXplore – High-Quality, Dispersible Graphenes

- High quality, low defects, easily dispersible graphene with controllable dimensions (2-5 layers; 100 nm – 20 μ m lateral)
- Multiple grades of graphene products (C, D, SD, E, F1, F2) for different applications
- Environmentally friendly processes
- Enable customers to develop ground-breaking products with a strong competitive advantage

Applications: thermal management, mechanical enhancement, and coatings, printing materials, batteries....



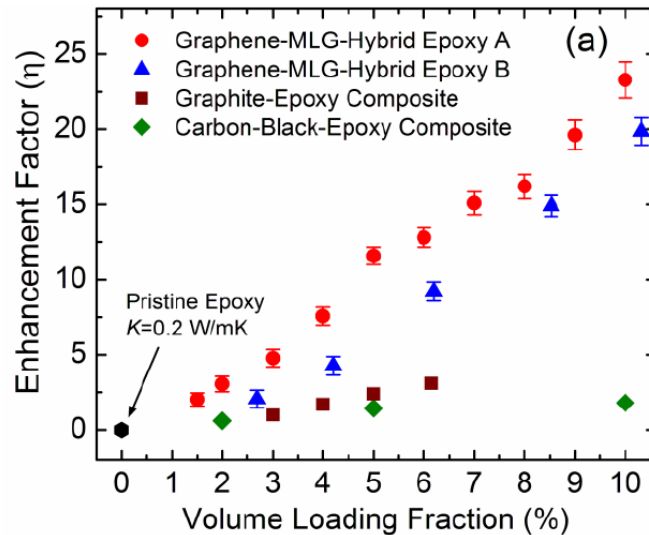
Graphene-Based Thermally Conductive Polymer Composites

Typical approaches for improving thermal conductivity:

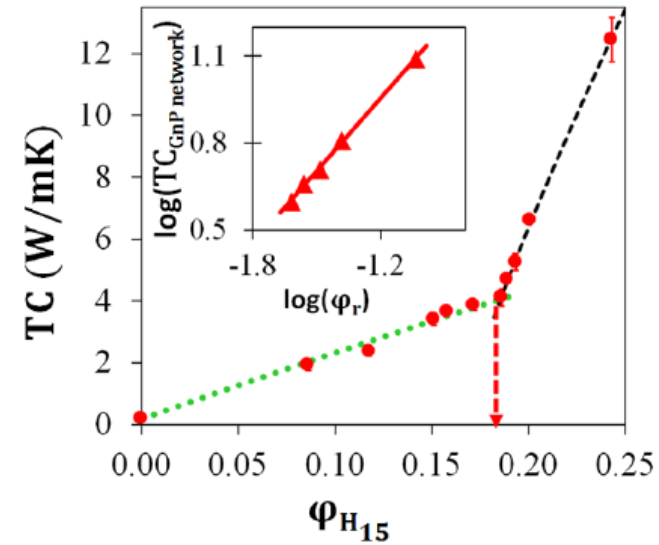
- Additives/filler: e.g., carbon fiber (0.5 W/m-k at 30 wt.%); silver and ceramic additives (1-5 W/m-k at ≤ 50 vol%)
- Orientation/Crystallization: PAI, PBI ~ 0.5 W/m-k; PT > 4.4 W/m-k; PE, PBO fiber > 40 W/m-k;
- Conjugation: Polyaniline > 0.5 W/m-k

Graphene can significantly enhance thermal conductivity of polymers at loadings of < 10 vol.%

Graphene-Based Thermally Conductive Polymer Composites



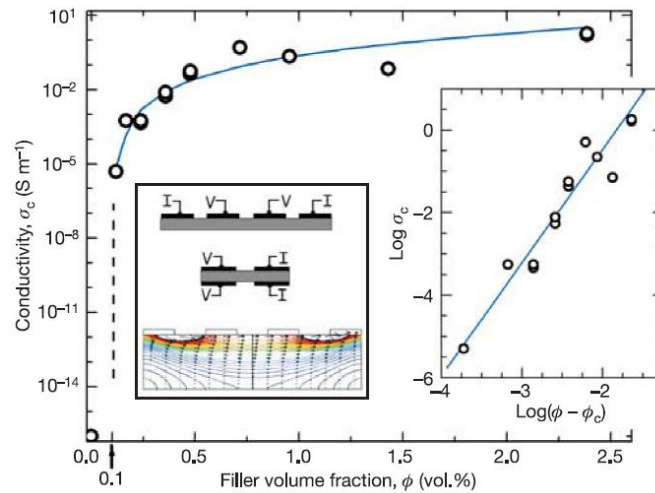
Shahil et al. Nano Lett, 2012, 12, 861



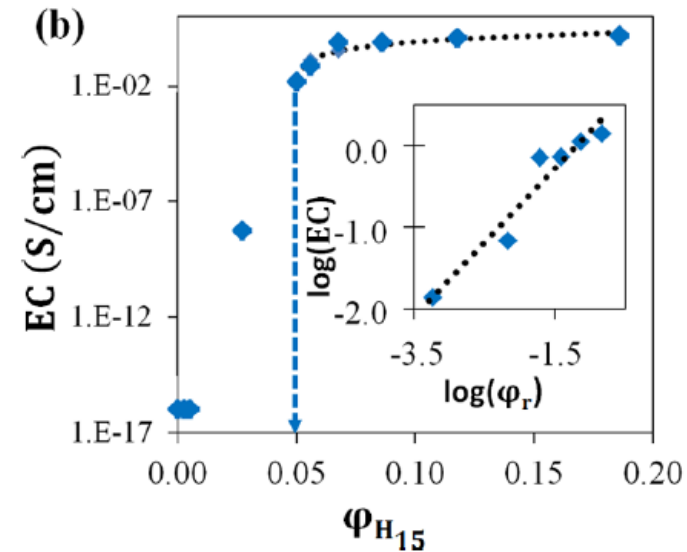
Shtein et al. Chem. Mater., 2015, 27, 2100

- Multilayer graphene showed significantly higher enhancement factor than carbon black and graphite
- A percolation concentration was found to be around 0.17 vol% for GnPs in epoxy

Graphene-Enabled Thermal conductivity vs Electrical Conductivity



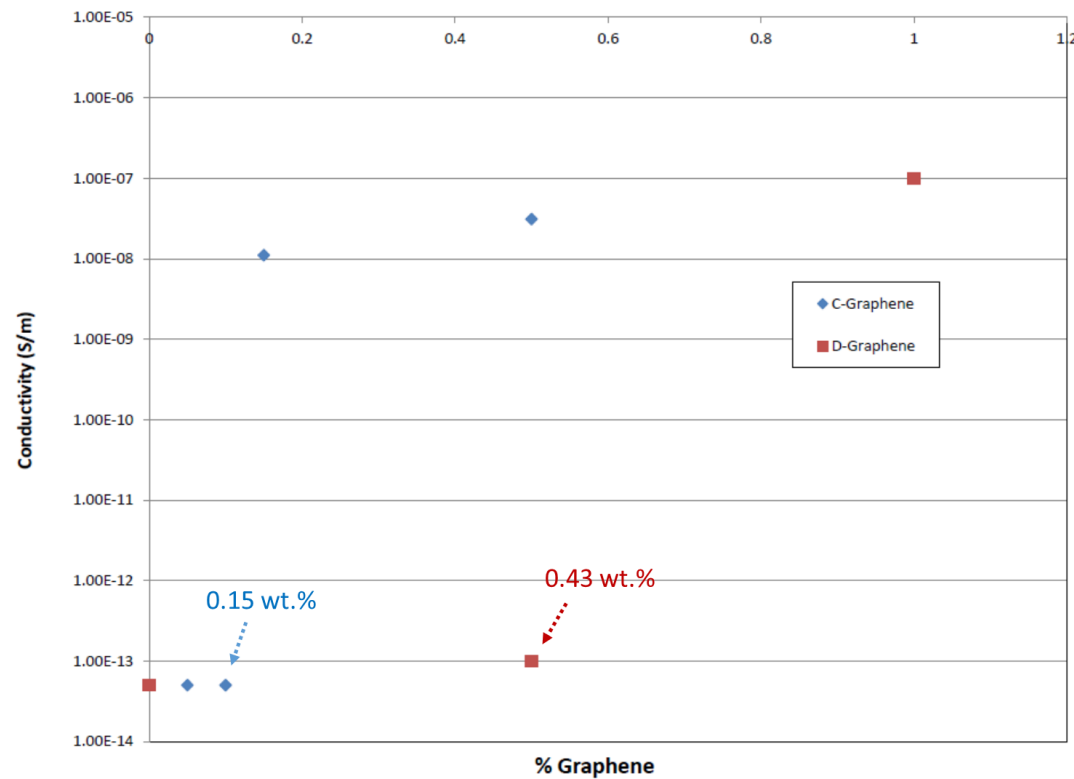
Stankovich et al. Nature, 2006, 442, 282



Shtein et al. Chem. Mater., 2015, 27, 2100

- Percolation concentration for electrical conductivity was found to be lower than that for thermal conductivity
- There is still limited polymeric materials with high thermal conductivity, as well as very low electrical conductivity and good processability (e.g., extrudable and injection moldable)

Electrical Percolation Concentration in Polymer Matrices



- NanoXplore graphene can be easily dispersed in thermoplastic/thermoset polymeric systems to give graphene/polymer composites
- Electrical percolation thresholds were found to be in range of 0.15 -0.5 wt.%
- It would be ideal if thermally conductive polymers could be produced at a low loading (e.g., <0.1 wt.%) of graphene

Development of Graphene-Based Thermally Conductive Polymers

- NanoXplore's graphene is compatible with a wide variety of polymers;
- A small loading of graphene into these polymer systems results into significantly improved properties, such as thermal, electrical, and mechanical properties

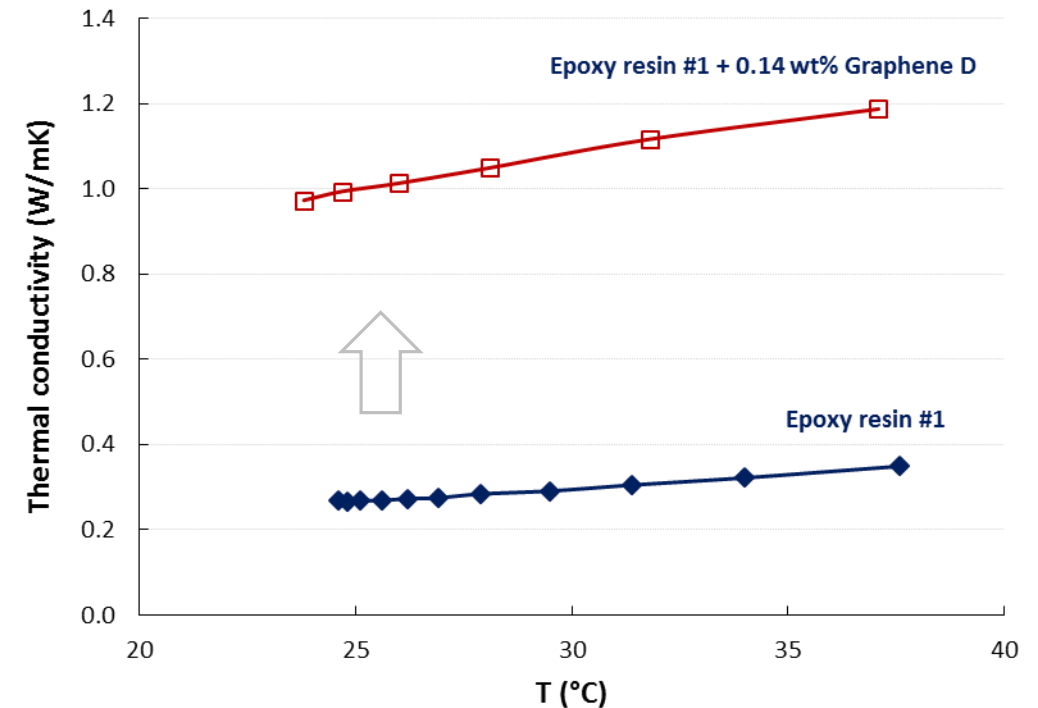
- Epoxies
- Polyurethane
- ABS
- PLA
- PE
- Silicone rubber
- PEI
- PC

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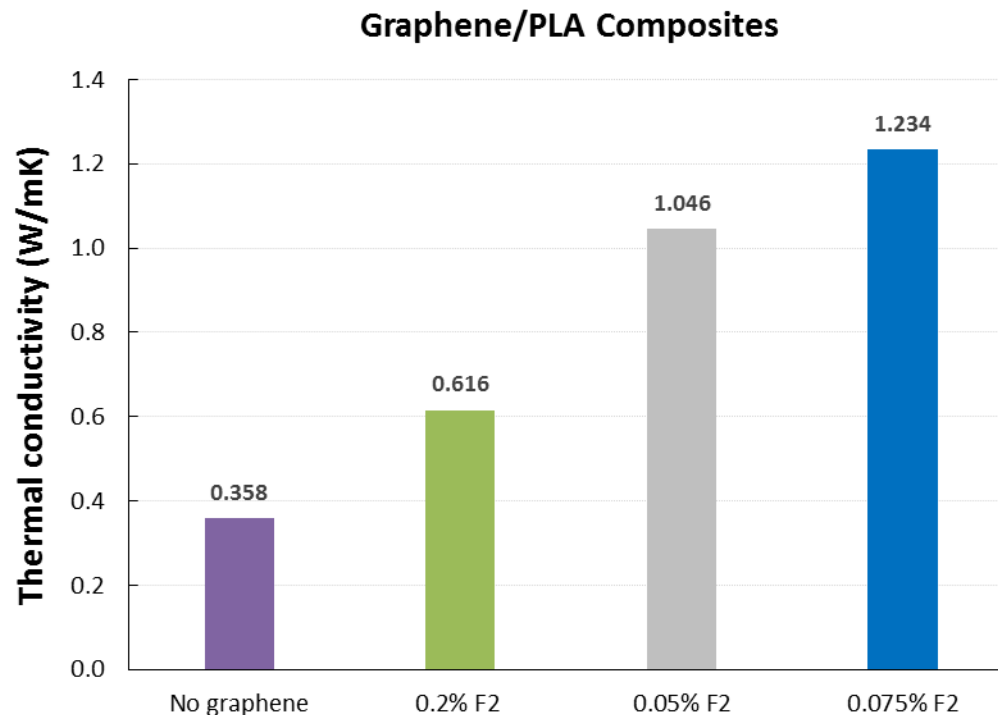


Graphene-Enhanced Thermally Conductive Epoxies

- Epoxy resins typically have thermal conductivity in range of 0.2 – 0.35
- 5-fold increase in thermal conductivity (>1 W/m-k) was achieved at 0.14 wt.% loading
- No substantial impact on electrical conductivity was found

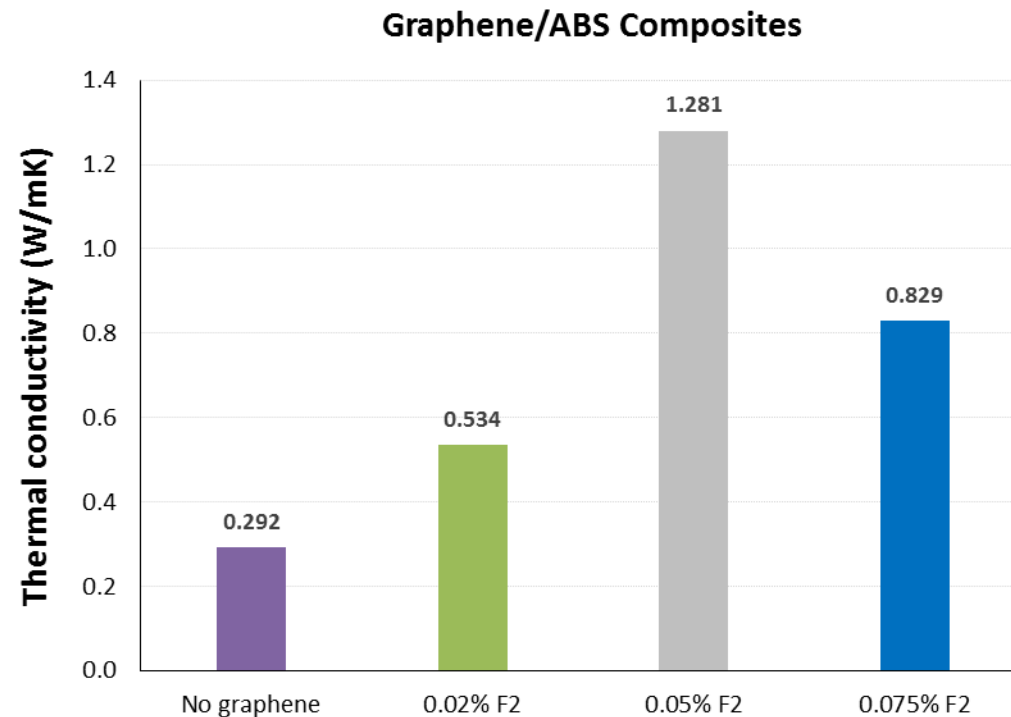


Graphene-Based Thermally Conductive Polymers



- Different grades of PLA were investigated for their thermal properties
- Thermal conductivity can be increased by 4-fold (>1.2 W/m-k) at very low loadings of <0.1 wt.%
- High-quality pellets and filaments of graphene/PLA can be easily produced through extrusion
- Packaging and printing applications are explored

Graphene-Based Thermally Conductive Polymers



- Graphene/ABS composites showed high mechanical strength
- Significantly improved thermal conductivity (>1.2 W/m-k) at low loadings of graphene (e.g., 0.05 wt.%)
- Readily extruded into high-quality filaments and injection mold into shaped articles
- Good 3D printability

Conclusions:

- NanoXplore has established large-scale production of ultra-high quality, dispersible graphene with a variety of different structural characteristics for different applications
- Graphene/polymer composites with very low graphene loading (e.g., ≤ 0.1 wt.%) were successfully prepared to enable high thermal conductivity (e.g., >1 W/m-k), low electrical conductivity, and good processability.
- Incorporation of low loading of graphene were also found effective in improving mechanical properties, hydrophobicity, and anti-corrosion properties



Thanks!

For more information come and meet us at the Booth L22!